Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/29/2025
PI Name:	Weaver, Ashley Ph.D.		
Project Title:	Development of Injury Assessment Reference Values for the Lumbar Spine Applicable to Spaceflight Use Cases		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **Dynamic Loads**:Risk of In-Mission Injury and Performance Decrements and Long-term Health Effects due to Dynamic Loads		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27101-4101	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	07/01/2024	**End Date:**	06/30/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Gayzik, Francis Ph.D. (Wake Forest University) Hsu, Fang-Chi Ph.D. (Wake Forest University) Stitzel, Joel Ph.D. (Wake Forest University) Devane, Karan Ph.D. (Wake Forest University)		
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Task Description:	Injury prediction and prevention in domains analogous to spaceflight dynamic events commonly rely on the use of anthropomorphic test devices (ATD), also known as crash test dummies. While ATDs have been effective in reducing injury in several domains, their use is limited in spaceflight as the dynamic events often involve the use of a seat or spacesuit that may not be compatible with the ATDs, and the dynamic events themselves may not be reproducible in a laboratory setting. Finite element human body models, on the other hand, can be outfitted with a spacesuit and fit into any seat design that may be used. These models can then predict the loads, accelerations, etc., of the body during the dynamic phases of flight. However, the acceptable level of injury risk in spaceflight is much lower than in analogous domains such as the automotive industry. There are currently no established thresholds for which loads on these human body models correspond to these lower levels of injury probability. This task seeks to identify these thresholds, also known as injury assessment reference values (IARVs), for the lumbar spine. This process starts by first identifying spaceflight relevant datasets of injurious and non-injurious loading to the lumbar spine. Next, the loading conditions from these datasets will be reconstructed with the human body models, and various outputs such as forces, bending moments, and/or stresses will be extracted. These outputs, also known as injury metrics, will then be correlated with the injury data to create injury risk curves. Finally, IARVs will be established, corresponding to the NASA definition of acceptable risk. This will enable more precise prediction of lumbar spine injury in a wider range of loading than is currently possible. This is a vital step in reducing injury risk during spaceflight.
Rationale for HRP Directed Research:	This task is NASA-directed due to research activities that are time sensitive, and these activities are highly constrained in that they involve working directly with multiple internal NASA programs, including the Human Landing System (HLS) and Orion. The research focuses on defining how the loads a crewmember may experience during unique events such as a lunar landing or surface operations correspond to a certain risk of injury, and it is expected to involve the discussion of proprietary information related to provider vehicles. The Human Physiology, Performance, Protection, and Operations Laboratory (H-3PO) at Johnson Space Center (JSC) has performed research in this area and has working relationships with subject matter experts at Wake Forest, which will make the effort more feasible and acceptable, and as cost-efficient as possible.
Research Impact/Earth Benefits:	Determining the likelihood of lower back injury to astronauts in launch and landing conditions will aid in the improvement of safety to the astronauts as well as vehicle occupants in lower energy collisions. Predicting risk and designing to prevent even minimal back injury is critical to spaceflight due to limited access to medical care which may be similar to what residents of rural areas experience. In such cases, availability of medical care may not reach accidents quickly enough to maximize the health outcomes for the occupants / patients. Therefore, injury risk analysis and designing for injury prevention is one of the most effective tools to improve quality of life.
Task Progress:	Phase 1: Literature review (Complete) - Datasets from studies with low-energy impacts on post-mortem human subjects (PMHS), human volunteers, and terrestrial analogs such as sports injuries or automotive impacts were collected and reviewed. The collected data was itemized by injury/non-injury and included injury severity on the abbreviated injury scale (AIS), location, anatomical structure, and mechanism of injury. Boundary conditions and validation data from these experimental studies have been documented to be used in finite element model recreations. A total of 88 lumbar spine minor and moderate injuries were counted. The most common lumbar spine anatomical structures to be injured in these conditions were vertebral bodies (n=50), transverse processes (n=30), and spinous processes (n=4). Phase 2: Finite Element Reconstruction (Ongoing) – Experimental conditions from studies identified in Phase 1 have been fully recreated for 5 studies, and 1 study is currently in the process of being reconstructed. The Global Human Body Models Consortium (GHBMC) average male simplified occupant model with a deformable spine (M50-OS+DeformSpine) finite element human body model has been adapted and validated for low-level loading conditions relevant to NASA's risk thresholds. Simulated instrumentation targeting commonly injured vertebral anatomy and injury mechanisms have been created. A total of 10 new lumbar spine virtual instruments have been added to the GHBMC model bringing the total number to 47 validated instruments. New lumbar instruments targeted intervertebral disc, spinous process, and facet cross-sectional forces and motion. CORrelation and Analysis (CORA) is the main technique for objective evaluation of the performance of the modeling reconstructions, including validation of lumbar spine kinetics and kinematics. CORA scores incorporate two main methods of evaluation, cross-correlation rating and corridor rating. The first method, cross-correlation, is an average of the comparison of magnitude, shape, and phase shift between the experimental data and simulated outputs. The second method, corridor rating, is an objective measure of how well a simulated trace falls between the corridors created by one and two standard deviations in experimental data.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Blade S, Mumtaz M, Gayzik FS, Stitzel J, Devane K, Weaver AA. "Development of injury assessment reference values for the lumbar spine applicable to spaceflight use cases." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025